Practica :

$$I = \int x^2 \cdot e^x \cdot dx$$

$$u = x^2 \qquad du = 2x$$

$$dv = e^x dx \Rightarrow \qquad v = e^x$$

$$\Rightarrow I = x^2 \cdot e^x - \underbrace{\int 2x \cdot e^x \, dx}_{J}$$

For J, we also do integration by parts

$$\begin{cases} u = 2x \\ dv = e^x dx \end{cases} \Rightarrow \begin{cases} du = 2 \\ v = e^x \end{cases}$$

$$J = 2x \cdot e^x - \int 2 \cdot e^x \, dx$$

$$= 2x \cdot e^x - 2e^x + C$$

$$\Rightarrow I = x^2 \cdot e^x - (2x \cdot e^x - 2e^x) + C$$

$$I = x^2 \cdot e^x - 2x e^x + 2e^x + C$$

(*) Tabular method for Integration by parts.

Repeated differentiation (u)	Repeated Integration (dv)
x^2	e^x
$2x$	e^x
2	e^x
0	e^x

$$I = x^2 \cdot e^x - 2x \cdot e^x + 2e^x + C$$

Integrals of the form

$$\int p(x) f(x)\, dx$$

Tabular Integration by Parts

Step 1. Differentiate $p(x)$ repeatedly until you obtain 0, and list the results in the first column.

Step 2. Integrate $f(x)$ repeatedly and list the results in the second column.

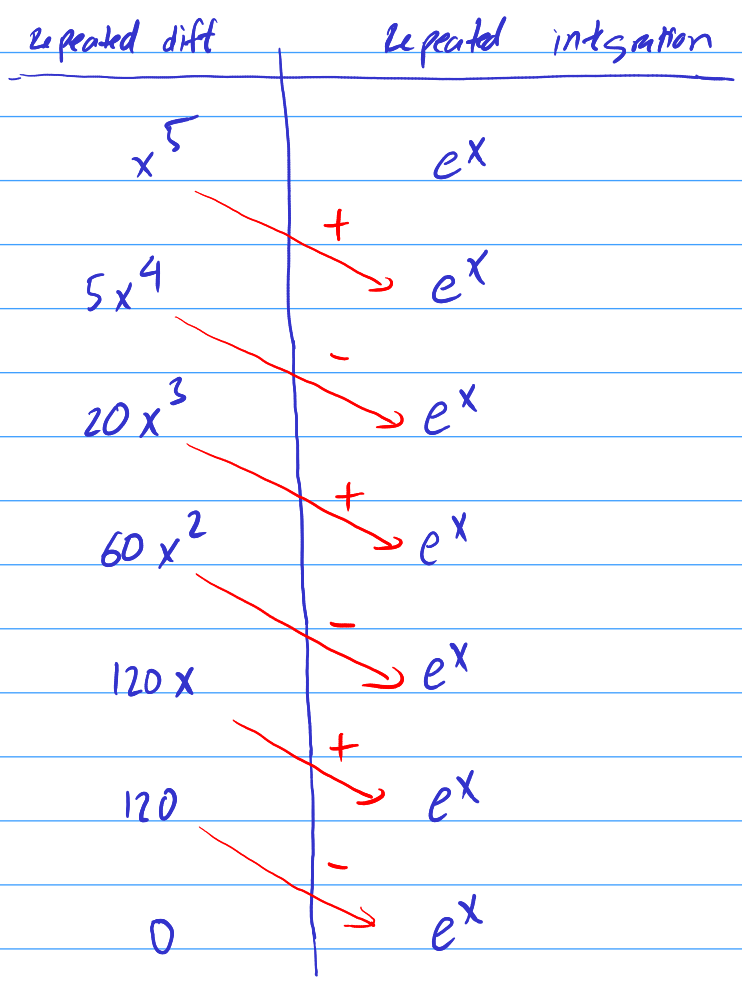
Step 3. Draw an arrow from each entry in the first column to the entry that is one row down in the second column.

Step 4. Label the arrows with alternating $+$ and $-$ signs, starting with a $+$.

Step 5. For each arrow, form the product of the expressions at its tip and tail and then multiply that product by $+1$ or -1 in accordance with the sign on the arrow. Add the results to obtain the value of the integral.

Example :

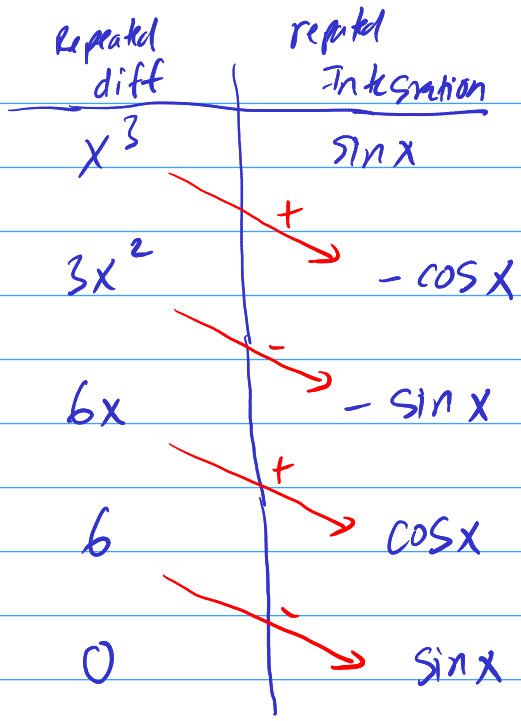
$$I = \int x^5 \cdot e^x \, dx$$



$$I = x^5 \cdot e^x - 5x^4 e^x + 20x^3 \cdot e^x - 60x^2 e^x + 120x e^x - 120 e^x + C$$

Example :

$$I = \int x^3 \sin x \cdot dx$$



$$I = -x^3 \cdot \cos x + 3x^2 \sin x + 6x \cos x - 6 \sin x + C$$

Example :

$$I = \int e^x \cdot \cos x \, dx$$

$$u = e^x \qquad du = e^x$$

$$dv = \cos x \, dx \qquad v = \sin x$$

$$\Rightarrow I = e^x \cdot \sin x - \underbrace{\int e^x \cdot \sin x \, dx}_{J}$$

For J: $\begin{cases} u = e^x \\ dv = \sin x\, dx \end{cases} \Rightarrow \begin{cases} du = e^x\, dx \\ v = -\cos x \end{cases}$

$$\Rightarrow J = -e^x \cdot \cos x - \int -(\cos x) \cdot e^x\, dx$$

$$J = -e^x \cos x + \int e^x \cos x \cdot dx$$

$$\Rightarrow J = -e^x \cos x + I$$

we have:

$$I = e^x \cdot \sin x - J$$

$$\Rightarrow I = e^x \cdot \sin x - \left(-e^x \cos x + I\right)$$

$$I = e^x \sin x + e^x \cos x - I$$

$$2I = e^x \sin x + e^x \cos x$$

$$I = \frac{1}{2}\left(e^x \sin x + e^x \cos x\right) + C$$

Example :

$$I = \int \sin^{-1} x \cdot dx$$

$$\begin{cases} u = \sin^{-1} x \\ dv = dx \end{cases} \Rightarrow \begin{cases} du = \frac{1}{\sqrt{1-x^2}} dx \\ v = x \end{cases}$$

$$I = x \cdot \sin^{-1} x - \int x \cdot \frac{1}{\sqrt{1-x^2}} dx$$

$$I = x \cdot \sin^{-1} x - \underbrace{\int \frac{x \cdot dx}{\sqrt{1-x^2}}}_{J}$$

Let $t = 1 - x^2$

$\Rightarrow dt = (1 - x^2)' \cdot dx$

$\Rightarrow dt = -2x\, dx$

$\Rightarrow \frac{dt}{-2} = x\, dx$

$\Rightarrow J = \int \frac{x\, dx}{\sqrt{1-x^2}} = \int \frac{dt/-2}{\sqrt{t}}$

$$= -\frac{1}{2}\int \frac{1}{t^{1/2}}\,dt$$

$$= -\frac{1}{2}\int t^{-1/2}\,dt$$

$$= -\frac{1}{2}\,\frac{t^{-1/2+1}}{-1/2+1} = -\sqrt{t}$$

$$J = -\sqrt{1-x^2} + C$$

$$\Rightarrow I = x\cdot \sin^{-1}x + \sqrt{1-x^2} + C$$